IPW3 : NACA0012 Code-to-Code Comparisons

POLYTECHNIQUE MONTRÉAL

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IPW3 – NACA0012 Participants

ID	Participant / Organization	Plot Color
001	CIRA	
002	AeroTex	
003	NRC	
004	ONERA	
006	Dassault	
007	Polytechnique Montréal	
008	Collins Aerospace	
009	Sikorsky	
010	NASA	
013	Boeing	
014	Airbus	
015	NIAR	
019	Synopsys	
020	Bombardier	

Geometry and Aerodynamic Conditions

NACA0012 – Wind-Tunnel View

Geometry

Parameter	Value
Airfoil	NACA0012
Chord, c [m]	0.5334
Angle of attack, α [°]	4.0

Aerodynamic Conditions

Parameter	Value
M_∞ [−]	0.3114
U_∞ [m/s]	102.15
T_∞ [K]	267.75
ρ_∞ [kg/m^3]	1.1884
Re_c [-]	3.8441×10^6

Quantities of Interest

$C_L, \quad C_D, \quad C_M, \quad C_p$

Icing Conditions

NACA0012 – Mesh Tunnel View

Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	AE3933
T_0 [°C]	-5.6	-2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m^3]	0.55	0.55
MVD [μm]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	**101.0**	**85.9**

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g).

Droplet Distribution: 1-, 3-, 7-, and 15-bin distributions with preserved total LWC, evaluated across all four grid levels.

Quantities of Interest

$\beta, \quad HTC, \quad T_s, \quad FF, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$

NACA0012 – Modeling Choices Across Submissions

Model / Method	001	002	004	007	008	009	014	019
Turbulence	k–ω TNT	k–ω SST	k–ω SST	SA	k–ω SST	k–ω SST	SA-neg	k–ω SST
Droplet formulation	Eul.	Lag.	Eul.	Eul.	Eul.	Eul.	Lag.	Eul.
HTC method	–	BL Int.	T_{rec}	T_{rec}	–	–	–	–
Thermodynamics	Mess.	Mess.	Mess.	Mess.	ICE3D/EID	Fluid film	Mess.	SWIM+
Geometry evolution	Lag./IB	Interp.	–	Lag.	ALE	–	Lag. disp.	Lag./remesh
No. of shots	1 / Multi	1	1	1	1 / Multi	Multi	–	Multi

Eul. = Eulerian; Lag. = Lagrangian; SA = Spalart–Allmaras; SA-neg = negative SA; BL Int. = boundary-layer integral; T_{rec} *= recovery temperature; Mess. = Messinger; EID = extended icing data; SWIM = shallow-water icing model; IB = immersed boundary; ALE = arbitrary Lagrangian–Eulerian; Interp. = interpolation; disp. = displacement; Multi = multilayer/multishot; – = not reported.*

Overall Trends

- k–ω SST and Eulerian droplet formulations are the predominant choices across submissions.
- Messinger-based thermodynamic approaches are widely used, while greater methodological diversity is observed in geometry evolution.
- Both single-shot and multilayer/multishot accretion strategies are represented, with substantial variation in the treatment of geometry updates.

NACA0012 – Summary of Submitted Data

AE3932

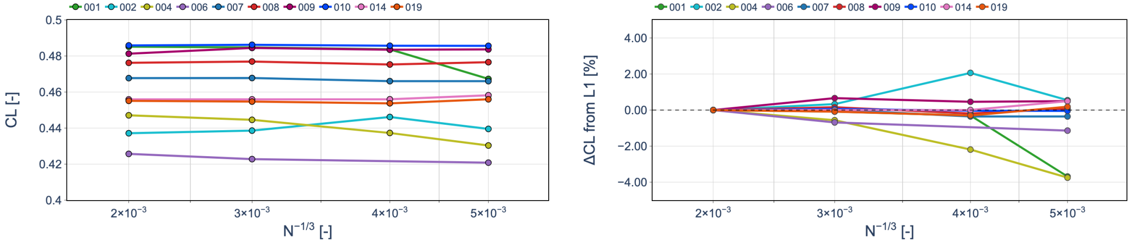
	L1				L2				L3				L4			
ID	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓				✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008		*				*				*				*		
009		*				*				*				*		
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020				✓				✓	✓	✓	✓	✓				✓
Total	**6**	**5**	**5**	**9**	**7**	**6**	**6**	**10**	**7**	**6**	**6**	**8**	**6**	**5**	**5**	**8**

AE3933

	L1				L2				L3				L4			
ID	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓	✓	✓	✓	✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
009	✓				✓				✓				✓			
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020												✓				
Total	**8**	**6**	**6**	**8**	**8**	**6**	**6**	**8**	**8**	**6**	**6**	**7**	**7**	**5**	**5**	**6**

✓: populated CutData received for the corresponding droplet-bin distribution.
∗: Excel results only, no CutData slices.

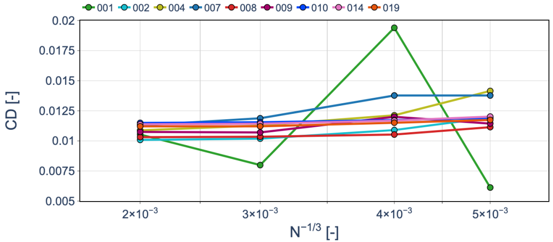
NACA0012 – Lift Coefficient Grid Convergence

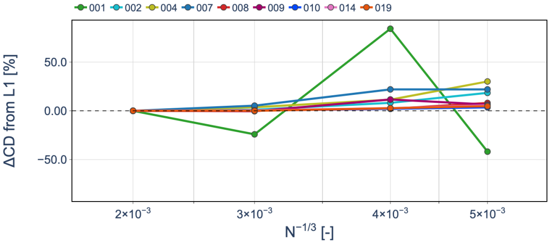


Observations

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit weak sensitivity to spatial grid refinement.
- Relative to L1, coarse-grid differences remain within approximately 4% across the submissions.

NACA0012 – Drag Coefficient Grid Convergence

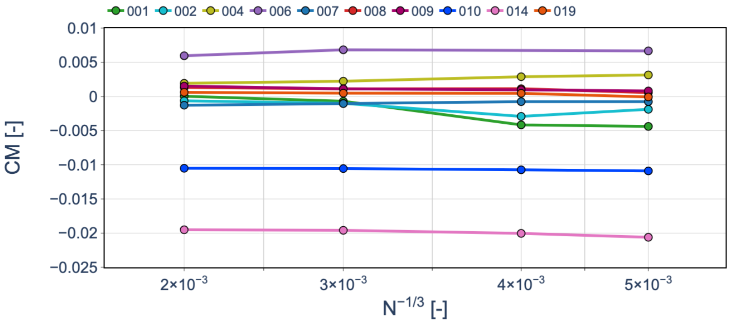




Observations

- At L1, the median drag coefficient is $C_D = 0.01088$ (**108.8 drag counts**), with an IQR of **7.57 drag counts** (7.0%).
- C_D exhibits the greatest sensitivity to spatial grid refinement among the integrated aerodynamic coefficients.
- On the coarser grids, most submissions deviate by up to approximately **30%** from their respective L1 values.
- One submission additionally exhibits pronounced non-monotonic behavior with grid refinement.

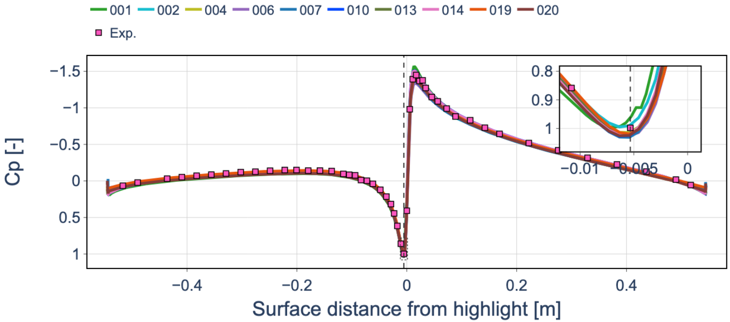
NACA0012 – Pitching Moment Grid Convergence



Observations

- At L1, the median pitching-moment coefficient is $C_M \approx 0$, with an IQR of 2.60×10^{-3}.
- Most submissions exhibit little variation in C_M across the grid hierarchy.

NACA0012 – Surface Pressure Distribution (L1)



Observations

- The L1 C_p distributions are tightly clustered and show good overall agreement with the experimental measurements.
- The largest code-to-code differences are localized near the leading-edge suction peak and stagnation region.
- Away from the leading edge, the predicted pressure distributions are nearly coincident across submissions.

NACA0012 – Aerodynamic Assessment Summary

L1 Code-to-Code Variability

	Median	IQR	Relative IQR
C_L	0.4677	0.0262	5.6%
C_D	108.8 counts	7.57 counts	7.0%
C_M	≈ 0	2.60×10^{-3}	–

Relative IQR is not reported for C_M because its median is close to zero.

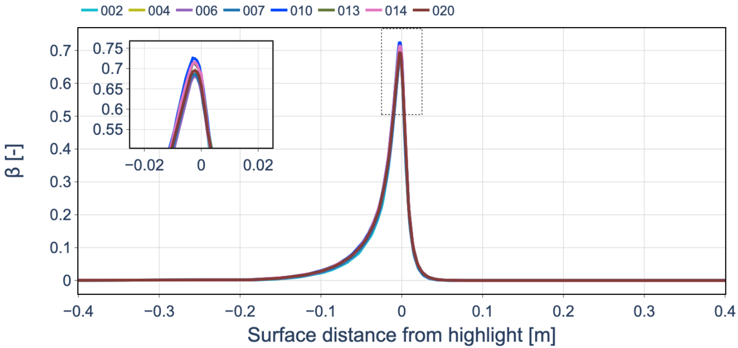
Grid-Refinement Behavior

- C_L and C_M exhibit limited within-code sensitivity to grid refinement for most submissions.
- C_D shows greater and less uniform grid sensitivity, including non-monotonic behavior for selected submissions.
- For most submissions, code-to-code differences exceed the changes produced by spatial refinement.

Aerodynamic Takeaway

Overall, the aerodynamic predictions are well clustered, with L1 C_p distributions showing good agreement with experiment despite differences in prescribed surface roughness. Among the integrated coefficients, C_D exhibits the greatest sensitivity to spatial grid refinement.

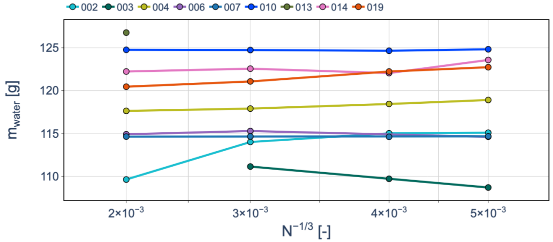
NACA0012 – Collection Efficiency (L1, 15 Bins)

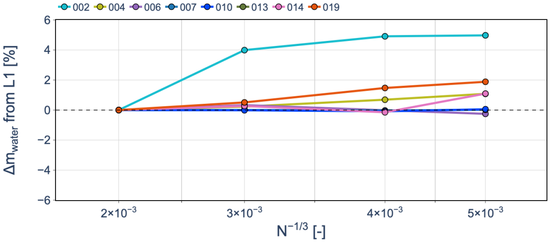


Observations

- The L1 collection-efficiency profiles are tightly clustered across submissions and exhibit very similar leading-edge behavior.
- Differences are concentrated near the peak region and impingement limits.

NACA0012 – Water Mass Grid Convergence (15 Bins)

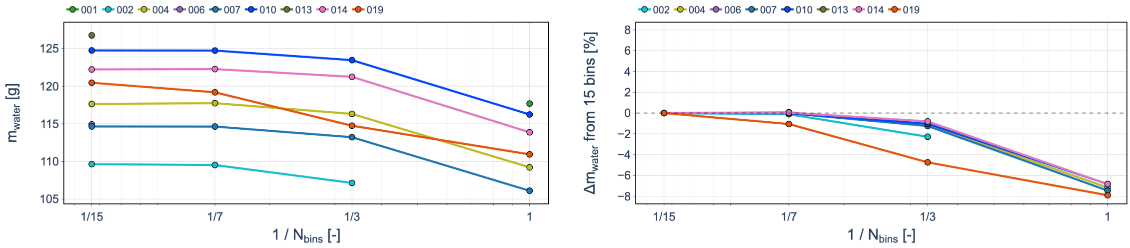




Observations

- For the 15-bin distribution, the L1 median impinged water mass is 119.05 g, with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity and remain close to their L1 prediction across the grid hierarchy.

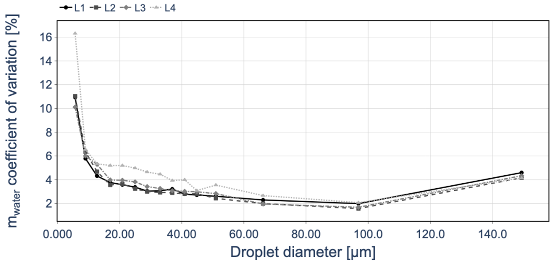
NACA0012 – Droplet Distribution Sensitivity (L1)



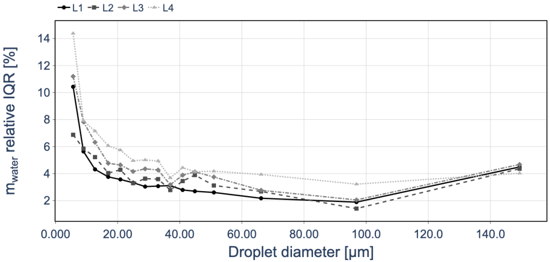
Observations

- The 15-bin and 7-bin distributions produce similar total impinged water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest additional change for most submissions.
- Relative to the 15-bin distribution, the single-bin approximation reduces the collected water mass by approximately 5–8% for most participants.

NACA0012 – Water-Mass Variability by Droplet Diameter (15 Bins)



Coefficient of Variation

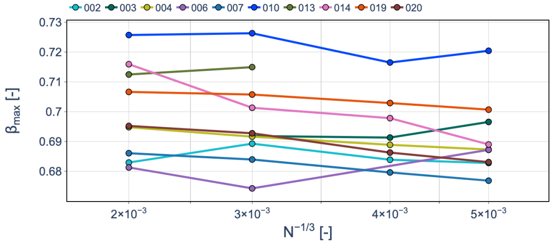


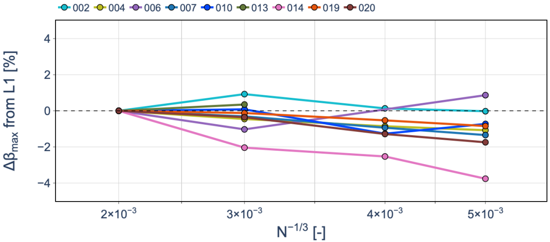
Relative IQR

Observations

- Code-to-code variability is strongly dependent on droplet diameter, with the largest spread observed for the smallest droplets
- Variability decreases rapidly with increasing diameter and remains comparatively low over most of the intermediate droplet-size range.
- The coarser L3–L4 grids generally exhibit greater variability than L1–L2, particularly for the smallest droplets.
- A slight increase in variability is observed again for the largest droplets.

NACA0012 – Peak Collection Efficiency Grid Convergence (15 Bins)

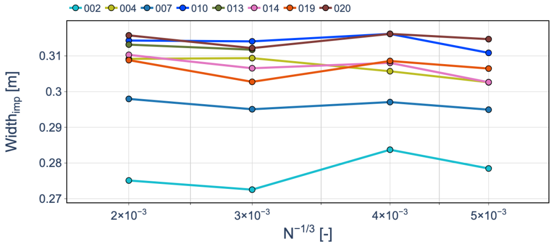


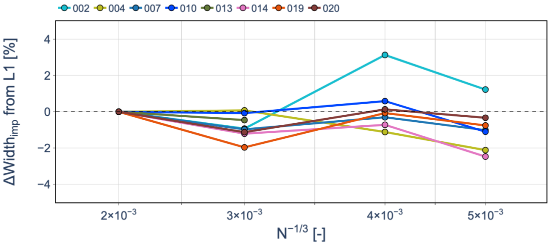


Observations

- At L1, the median peak collection efficiency is $\beta_{\text{max}} = 0.695$, with an IQR of 0.0209 (3.0%).
- Most submissions exhibit limited sensitivity of β_{max} to spatial grid refinement.
- Similar code-to-code clustering and grid-refinement behavior are observed for the other droplet-size distributions (not shown).
- The location of β_{max} was also examined (not shown); however, it is strongly influenced by the local leading-edge surface discretization. Consequently, no clear grid-convergence conclusion is drawn for the peak position.

NACA0012 – Impingement Width Grid Convergence (15 Bins)





Observations

- At L1, the median impingement width is $0.309\ \mathrm{m}$, with an IQR of $8.93 \times 10^{-3}\ \mathrm{m}$ (2.9%).
- The impingement width is not fully grid-converged, although the variations between successive grid levels remain relatively small for most submissions.
- The remaining grid dependence partly reflect the sensitivity of the impingement limits to the surface discretization.

NACA0012 – Droplet Impingement Assessment

Spatial Grid Convergence

- Water mass, β_{max}, and impingement width show limited grid sensitivity for most submissions.
- Code-to-code variability is generally comparable to or larger than the grid-refinement effect.
- Peak position was examined, but its sensitivity to the local leading-edge discretization prevented a clear convergence assessment.

Droplet-Size Effects

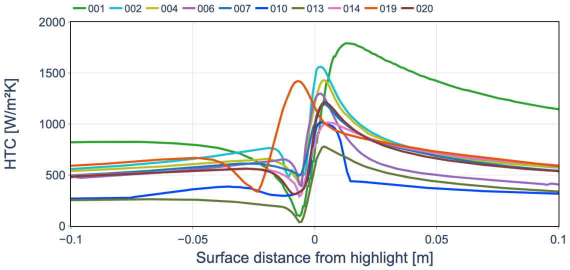
- The 7-bin distribution closely reproduces the 15-bin results, while the single-bin approximation systematically underpredicts collected water mass.
- Code-to-code variability is largest for the smallest droplets and decreases substantially over the intermediate diameter range.

Impingement Takeaway

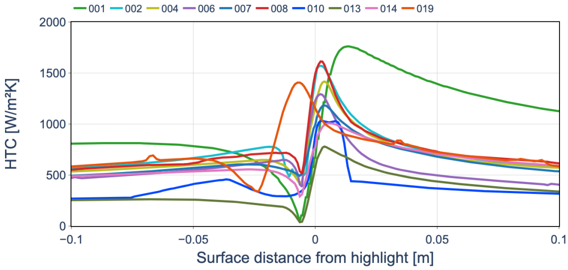
Overall, the impingement predictions are relatively consistent with spatial grid refinement. Droplet-size discretization has a clearer systematic influence, with 7 bins providing results close to the 15-bin reference.

Convective Heat Transfer & Thermodynamics

NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932

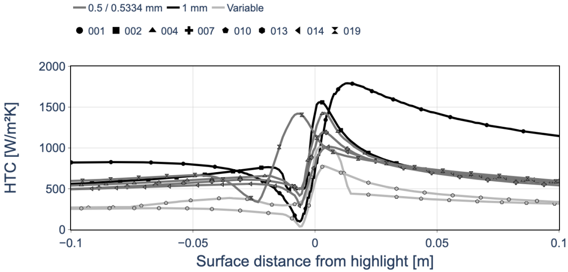


AE3933

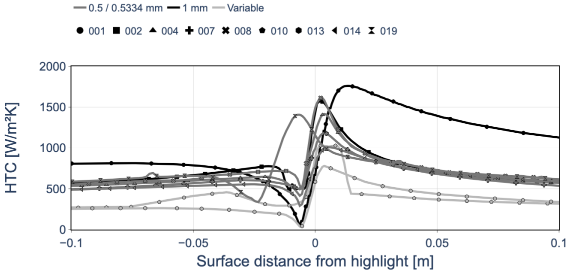
Observations

- Considerable code-to-code variability is observed in *HTC*, particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, while the remaining code-to-code variability must also be influenced by differences in the *HTC* computation methodology.

NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932 – Grouped by Roughness



AE3933 – Grouped by Roughness

Observations

- Considerable code-to-code variability is observed in *HTC*, particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, while the remaining code-to-code variability must also be influenced by differences in the *HTC* computation methodology.

NACA0012 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local *HTC* distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$\boxed{Q'_c = \int HTC(s)\,[T_s(s) - T_{\mathrm{rec}}(s)]\; ds} \qquad [Q'_c] = \mathrm{W/m}.$$

The submitted *HTC* and surface temperature T_s are used directly. The recovery temperature T_{rec}, however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

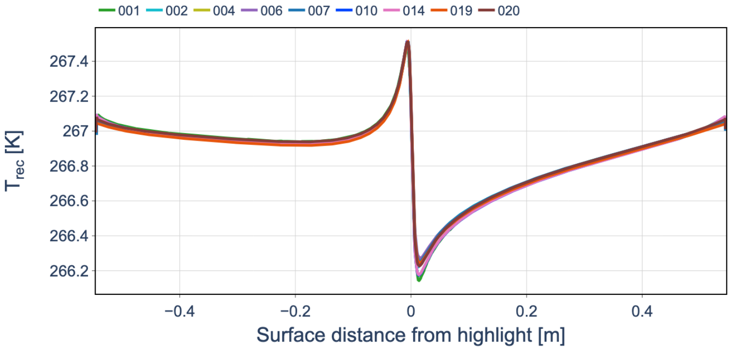
$$\boxed{C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{\mathrm{rec}}(s)}$$

Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}, \qquad M_e^2 = \frac{2}{\gamma - 1}\left[\left(\frac{p_0}{p_e}\right)^{(\gamma-1)/\gamma} - 1\right],$$

$$T_e = \frac{T_0}{1 + \dfrac{\gamma - 1}{2} M_e^2}, \qquad T_{\mathrm{rec}} = T_e\left[1 + r\frac{\gamma - 1}{2} M_e^2\right], \qquad r = Pr^{1/3} \approx 0.9.$$

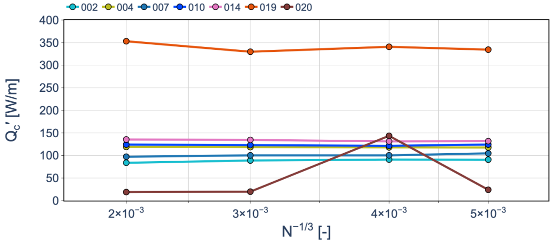
NACA0012 – Reconstructed Recovery Temperature (L1)



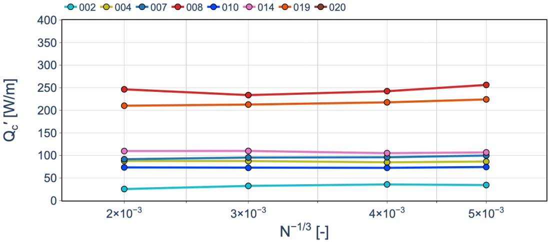
Observations

- The reconstructed T_{rec} distributions are closely clustered across submissions.
- This consistency follows directly from the close agreement previously observed in the submitted C_p distributions.
- Differences in Q_c' are therefore expected to originate primarily from HTC and T_s, rather than from the reconstructed T_{rec}.

NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932

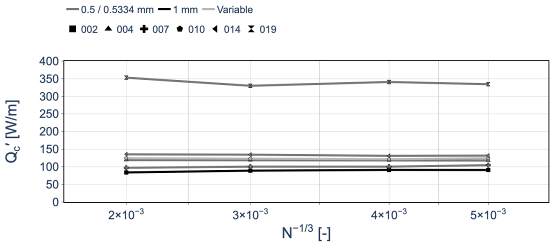


AE3933

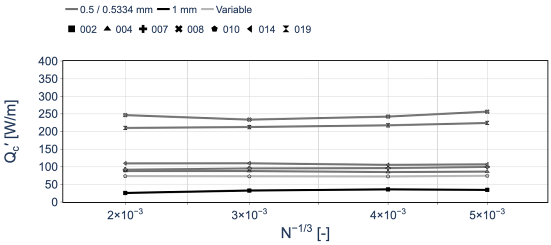
Observations

- Most submissions exhibit limited grid sensitivity in the integrated Q_c'.
- Grouping by prescribed roughness reveals a code-dependent roughness effect, while substantial variability remains within the roughness groups.

NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932 – Grouped by Roughness

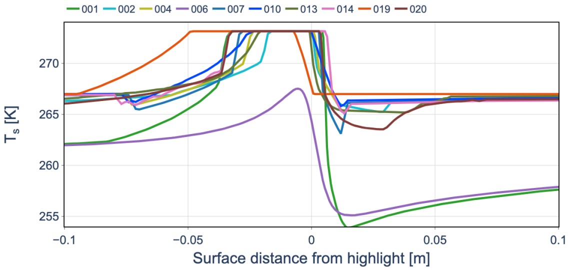


AE3933 – Grouped by Roughness

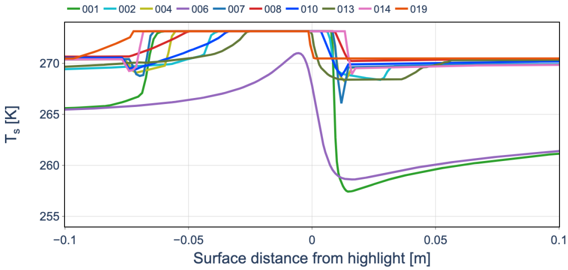
Observations

- Most submissions exhibit limited grid sensitivity in the integrated Q'_c.
- Grouping by prescribed roughness reveals a code-dependent roughness effect, while substantial variability remains within the roughness groups.

NACA0012 – Surface Temperature Comparison (L1)



AE3932

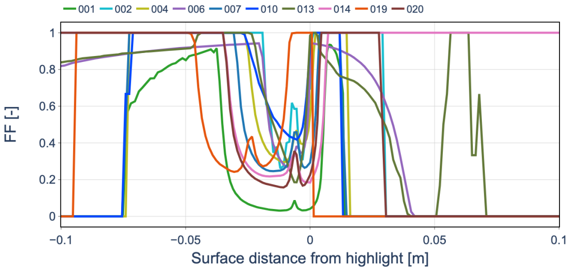


AE3933

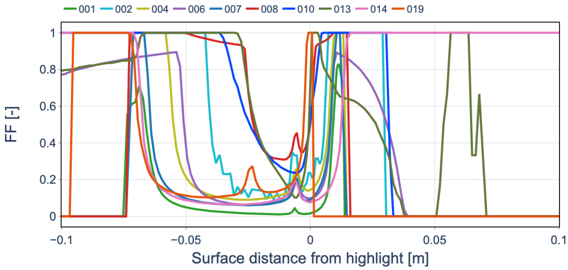
Observations

- Substantial code-to-code variability is observed in the surface-temperature distributions for both conditions.
- AE3933 exhibits a wider region where the surface temperature remains near the freezing temperature, consistent with its warmer freestream condition.
- The extent of this near-freezing region directly influences the predicted freezing-fraction distribution and, ultimately, the ice-accretion limits.

NACA0012 – Freezing Fraction Comparison (L1)



AE3932



AE3933

Observations

- Considerable code-to-code variability is observed in the freezing-fraction distributions for both conditions.
- AE3932 generally exhibits a narrower low-freezing-fraction region near the leading edge, whereas the warmer AE3933 condition tends to produce a wider region.
- This trend is consistent with the broader near-freezing surface-temperature region observed for AE3933.
- However, the large code-to-code variability and differences in transition locations make the freezing-fraction distributions difficult to interpret quantitatively.

NACA0012 – HTC & Thermodynamics Assessment

Convective Heat Transfer

- Local *HTC* exhibits substantial code-to-code variability, with prescribed roughness explaining only part of the spread.
- Reconstructed T_{rec} is closely clustered, consistent with the agreement previously observed in C_p.
- Integrated Q_c' exhibits limited grid sensitivity for most submissions.

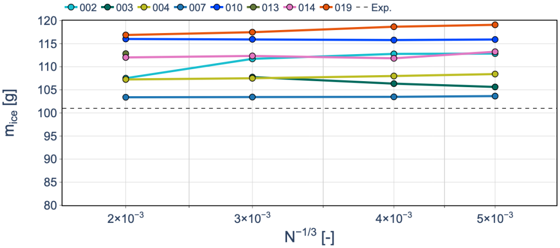
Thermodynamic Response

- Surface temperature and freezing fraction exhibit substantially greater code-to-code variability.
- The warmer AE3933 condition generally produces a wider near-freezing surface-temperature region and a correspondingly wider low-freezing-fraction region.

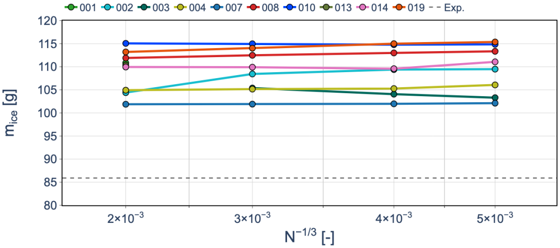
Thermodynamics Takeaway

Convective heat transfer is relatively insensitive to grid refinement but exhibits substantial code-to-code variability. The warmer AE3933 condition produces a broader near-freezing region, which is expected to influence the freezing behavior and subsequent ice-accretion limits.

NACA0012 – Ice Mass Grid Convergence (15 Bins)



AE3932

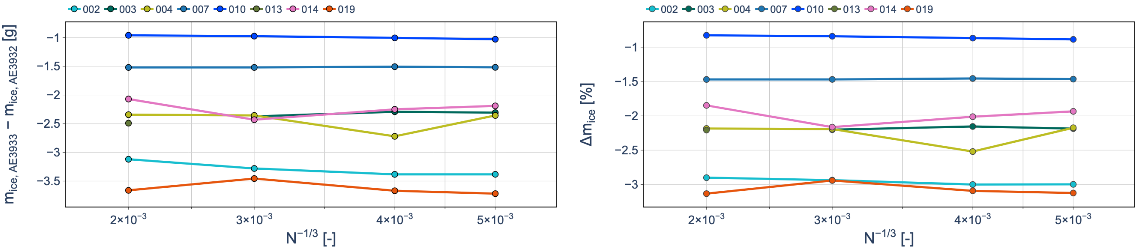


AE3933

Observations

- Most submissions exhibit limited sensitivity of the predicted ice mass to spatial grid refinement for the 15-bin distribution.
- AE3933 generally produces less ice mass than AE3932, with predicted reductions of at most approximately 3%, compared with the approximately 15% observed experimentally.

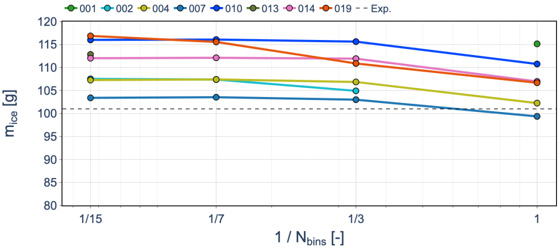
NACA0012 – Ice Mass Difference Between Conditions (15 Bins)



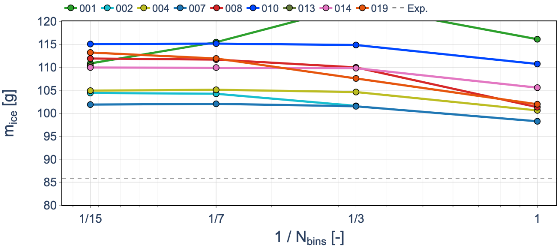
Observations

- The matched-participant comparison confirms systematically lower ice mass for AE3933 than AE3932 for most submissions.
- The numerical predictions reproduce the experimental trend of lower ice mass for AE3933, but predict a substantially smaller difference than the approximately 15% observed experimentally.

NACA0012 – Ice Mass Sensitivity to Droplet Distribution (L1)



AE3932

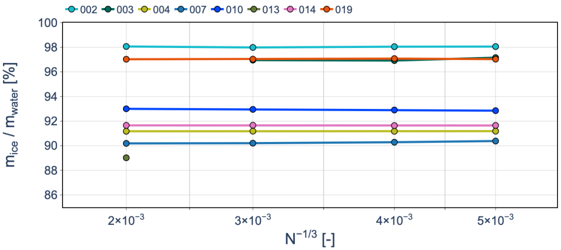


AE3933

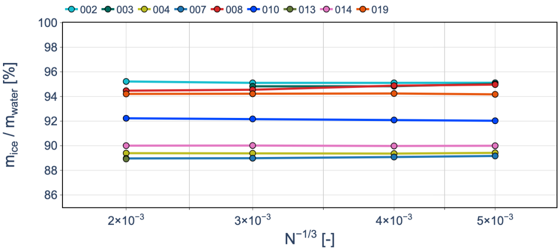
Observations

- The 15- and 7-bin distributions generally produce comparable ice masses for both conditions.
- Sensitivity increases when the distribution is reduced to 3 bins and becomes more pronounced for the single-bin representation.

NACA0012 – Ice-to-Water Mass Ratio



AE3932

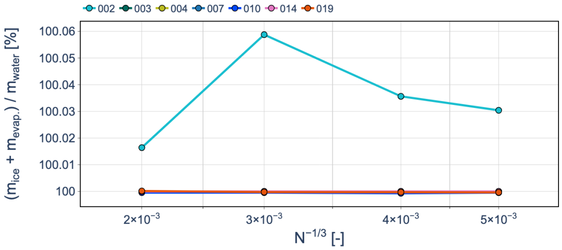


AE3933

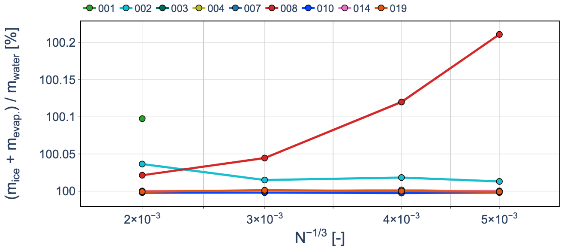
Observations

- Approximately 90% of the impinged water is converted into ice for both conditions.
- AE3933 consistently exhibits a lower ice-to-water ratio than AE3932, indicating greater water loss.
- Since the water impingement is nearly identical between the two conditions, this difference originates primarily from the thermodynamic mass balance.

NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932

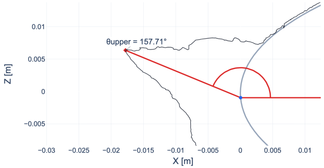


AE3933

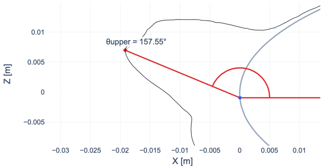
Observations

- The combined ice and evaporated-water mass accounts for approximately 100% of the impinged water for most submissions.
- This indicates that the lower ice-to-water ratio for AE3933 is primarily associated with increased evaporation.
- Values slightly above 100% are observed for a few submissions and are likely attributable to numerical integration or round-off errors.

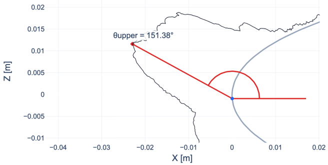
NACA0012 – Experimental Upper Horn Angle



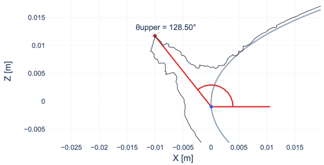
MinCCS



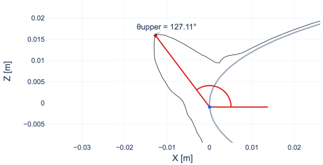
MeanCCS
AE3932



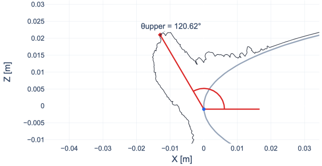
MaxCCS



MinCCS



MeanCCS
AE3933

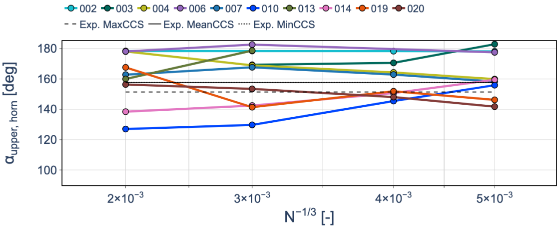


MaxCCS

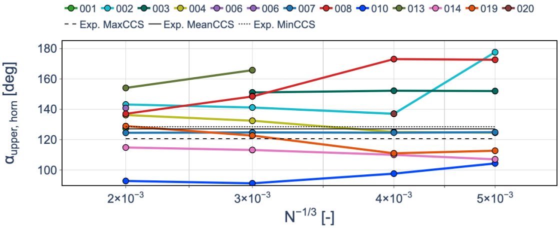
Experimental Reference

- The experimental upper-horn angle ranges from approximately 151–158° for AE3932 and 121–129° for AE3933.

NACA0012 – Upper Horn Angle Grid Convergence (15 Bins)



AE3932

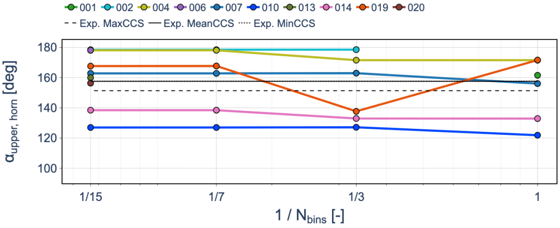


AE3933

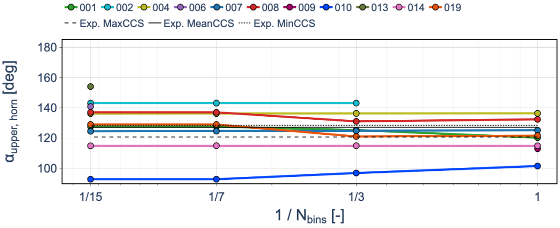
Observations

- Horn-angle predictions show substantial code-to-code variability with no systematic grid convergence.
- Grid refinement affects the spread differently between the two conditions.
- AE3933 consistently predicts a smaller upper-horn angle than AE3932, consistent with experiment.
- L1 scatter is comparable between conditions ($\mathrm{IQR} \approx 16\%$ and 13%).

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution (L1)



AE3932

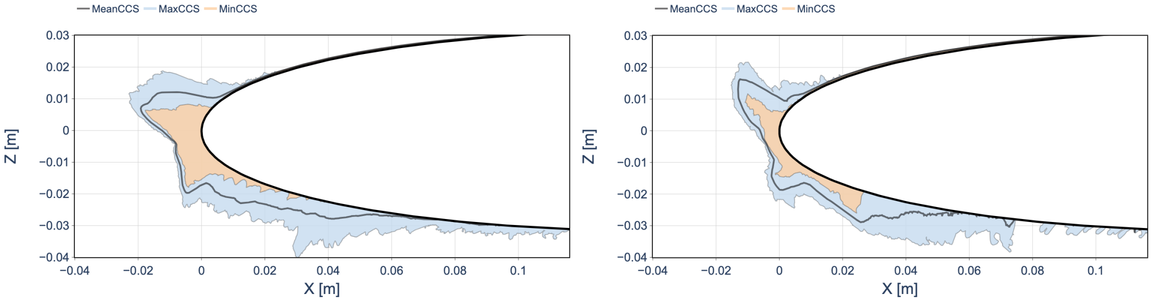


AE3933

Observations

- The upper-horn angle exhibits only small variations across the different droplet-size distributions for most submissions.
- The overall horn orientation is therefore relatively insensitive to the droplet-size discretization.

NACA0012 – Ice Shapes (L2, 15 Bins)

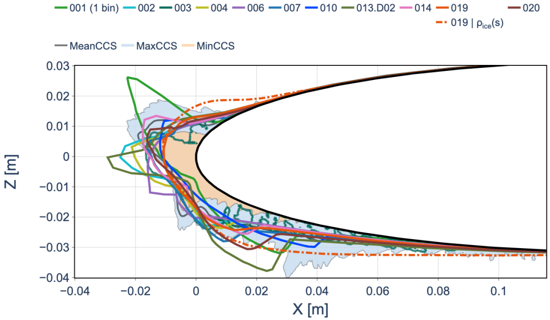


AE3932 – Experiment **AE3933 – Experiment**

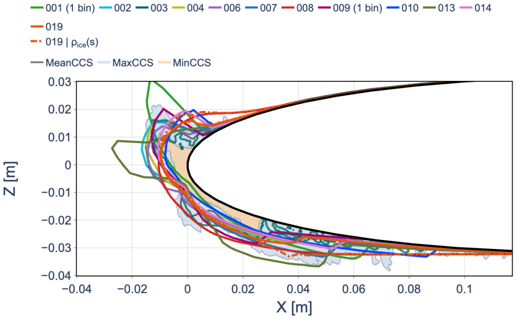
Observations

- Experiments show a clear difference in ice shape, particularly in upper-horn orientation.
- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
- The experimental difference between conditions is reproduced to varying degrees.
- Roughness effects are code-dependent.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932



AE3933

Observations

- Experiments show a clear difference in ice shape, particularly in upper-horn orientation.
- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
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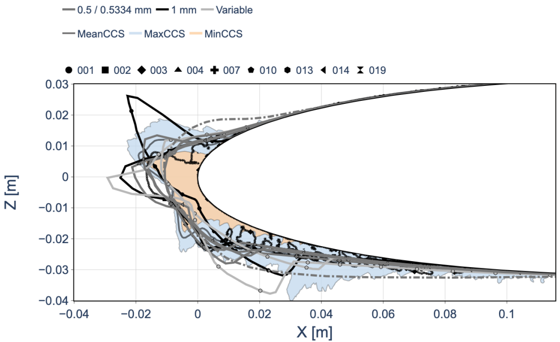
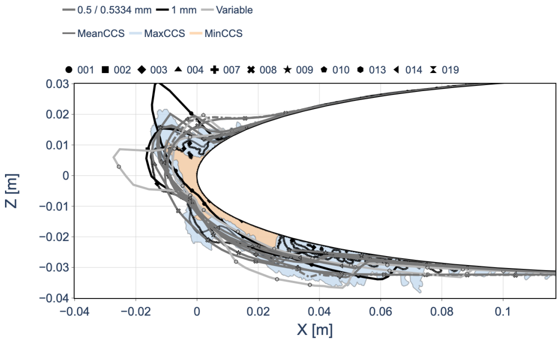
- Roughness effects are code-dependent.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Grouped by Roughness

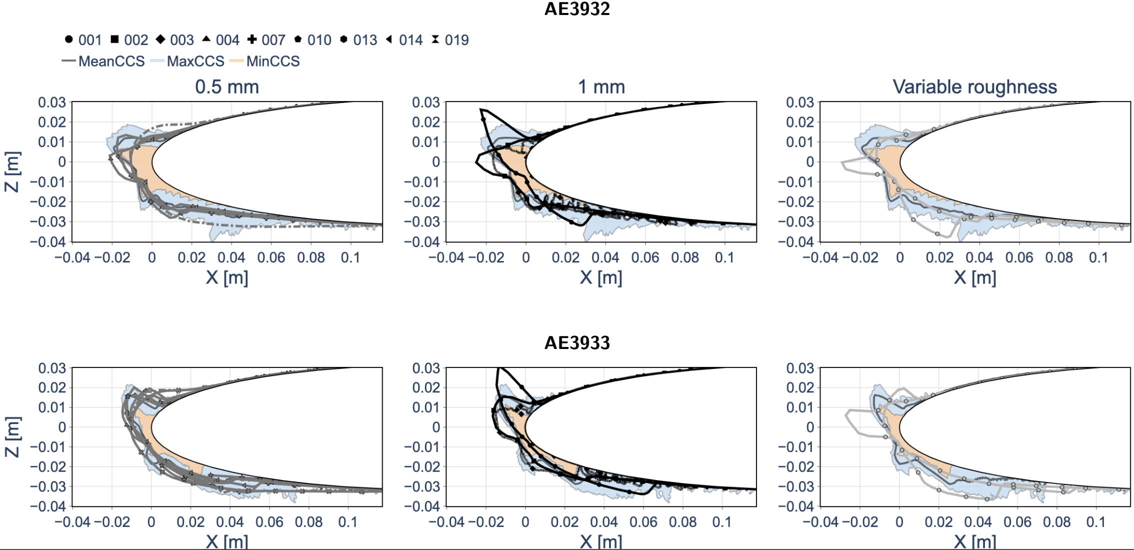


AE3933 – Grouped by Roughness

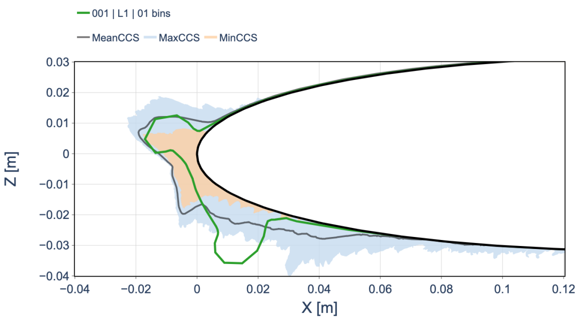
Observations

- Experiments show a clear difference in ice shape, particularly in upper-horn orientation.
- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
- The experimental difference between conditions is reproduced to varying degrees.
- Roughness effects are code-dependent.

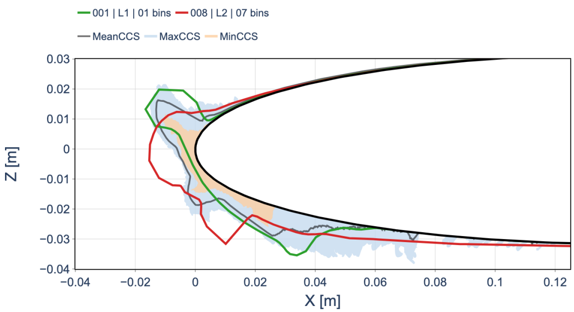
NACA0012 – Effect of Surface Roughness on Ice Shapes (L2, 15 Bins)



NACA0012 – Multilayer Ice-Shape Comparison



AE3932



AE3933

Observations

- Multilayer predictions suggest slight improvement, but the limited dataset prevents a conclusive assessment.

NACA0012 – Ice Accretion Summary

Verification

- Ice mass shows limited grid sensitivity, with an L1 code-to-code IQR of approximately 5% for both icing conditions; horn orientation exhibits greater variability, while the 15- and 7-bin distributions generally provide comparable results.

Condition-to-Condition Response

- AE3933 generally produces less ice than AE3932, although the predicted reduction remains substantially smaller than the approximately 15% observed experimentally.
- A lower upper-horn angle is consistently predicted for AE3933, reproducing the experimentally observed geometric shift between the two conditions.

Ice Geometry

- Considerable code-to-code variability remains in the predicted ice shapes, particularly in horn orientation and local geometry.
- The available multilayer predictions suggest improved agreement with experiment and capture additional geometric features, although the limited number of submissions prevents a conclusive assessment.

Ice Accretion Takeaway

The numerical predictions reproduce the experimental trend between AE3932 and AE3933, but underestimate its magnitude, while substantial variability remains in the final ice geometry.

NACA0012 – Overall Assessment

Overall Assessment

- **Aerodynamics:** Predictions are well clustered; C_p agrees well with experiment, while C_D shows the greatest grid sensitivity.
- **Droplet Impingement:** Limited grid sensitivity is observed, with comparable results between the 15- and 7-bin distributions.
- **HTC & Thermodynamics:** *HTC* exhibits substantial code-to-code variability, while AE3933 produces the expected broader near-freezing surface-temperature region.
- **Ice Accretion:** AE3933 predicts less ice and a smaller upper-horn angle than AE3932, reproducing the experimental trends, although the ice-mass reduction is substantially underestimated. Considerable variability remains in the final ice geometry.

Overall Takeaway

Aerodynamics and water impingement are comparatively consistent across submissions, while larger differences emerge through thermodynamics and final ice accretion. The experimental distinction between AE3932 and AE3933 is qualitatively captured, but its magnitude remains underpredicted.